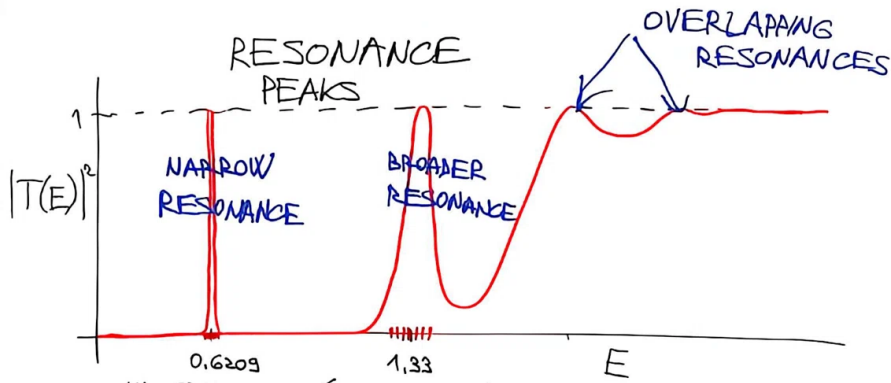


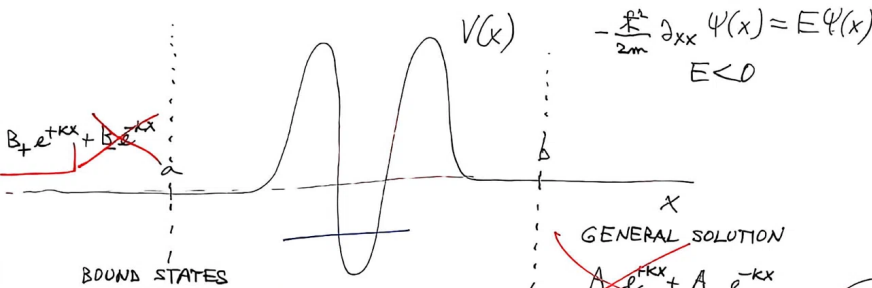
NONHERMITIAN QUANTUM MECHANICS

1926 TDSE

1930's GAMOW, SIEGERT



IN STANDARD (HERMITIAN) QM:
RESONANCES ARE SEEN SO FAR
AS OPAQUE FEATURES IN $|T(E)|^2$



$$-\frac{\hbar^2}{2m} \partial_{xx} \psi(x) = E \psi(x)$$

$E < 0$

$$k = \sqrt{-2mE} > 0$$

NHQM: IMPOSE (*) & (**) WHILE TAKING $z \in \mathbb{C}$

$$E = \frac{\hbar^2 z^2}{2m}$$

$$\left(-\frac{\hbar^2}{2m} \partial_{xx} + V(x) \right) \psi(x) = \frac{\hbar^2 z^2}{2m} \psi(x)$$

& (*) & (**)

$$z = z_R + i z_I$$

$$E = \underbrace{\frac{\hbar^2}{2m} (z_R^2 - z_I^2)}_{\text{Re } E} + i \underbrace{\frac{\hbar^2}{2m} 2 z_R z_I}_{\text{Im } E}$$

SIEGERT STATES

TEXTBOOKS:

$$\hat{H} = -\frac{\hbar^2}{2m} \partial_{xx} + V(x)$$

DOMAIN: $\mathbb{C}$ NHQM

$$\hat{H} = \hat{H}^\dagger \text{ HERMITIAN}$$

$$\left(-\frac{\hbar^2}{2m} \partial_{xx} + V(x) \right) \psi(x) = E \psi(x)$$

$\psi(x)$ IS SQUARE INTEGRABLE

BOUNDARY CONDITIONS:

$$\psi(x \rightarrow \pm\infty) \sim e^{\mp kx}$$

EQUIVALENTLY

$$\psi(x \rightarrow \pm\infty) \sim e^{\pm i z x}$$

$$z = i k$$

$$k = \sqrt{2mE}$$

$$(\partial_x - i z) \psi(x) \Big|_{x \geq b} = 0 \quad (*) \text{ SIEGERT BOUNDARY CONDITIONS}$$

$$(\partial_x + i z) \psi(x) \Big|_{x \leq a} = 0 \quad (**)$$

ASSUME E REAL

$$-\frac{\hbar^2}{2m} \psi''(x) + V(x) \psi(x) = E \psi(x) \quad / \cdot \psi^*(x)$$

$$-\frac{\hbar^2}{2m} \psi^{*''}(x) + V(x) \psi^*(x) = E \psi^*(x) \quad / \cdot \psi(x)$$

$$\frac{d}{dx} (\psi^*(x) \psi'(x) - \psi(x) \psi^{*'}(x)) = 0$$

$$\frac{d}{dx} (\psi^*(x) \psi'(x) - \psi(x) \psi^{*'}(x)) = 0$$

$$F(x) = \psi^*(x) \psi'(x) - \psi(x) \psi^{*'}(x)$$

IS x-DEPENDENT

$$\psi(x > b) = A e^{+i\lambda x} \quad \lambda > 0$$

$$\psi(x < a) = B e^{-i\lambda x}$$

$$\text{EVALUATE } F(x > b) = \dots$$

$$F(x < a) = \dots$$

$$F(x > b) = \cancel{A^* e^{-i\lambda x}} \cancel{A e^{+i\lambda x}} - \cancel{A^* e^{-i\lambda x}} \cancel{A e^{+i\lambda x}} = |A|^2 2i\lambda$$

$$F(x < a) = \dots = |A|^2 (-2i\lambda)$$

HERMITIAN QM

$$\langle \psi^{(1)} | \psi^{(2)} \rangle = \int_{-\infty}^{+\infty} dx \psi^{(1)*}(x) \psi^{(2)}(x)$$

* CAN BE OMITTED FOR BOUND STATES BECAUSE THE BOUND STATE ψ 'S ARE REAL

NHQM

ZEL'DOVICH

$$(\psi^{(1)} | \psi^{(2)}) = \lim_{\epsilon \rightarrow 0^+} \int_{-\infty}^{+\infty} dx \psi^{(1)}(x) \psi^{(2)}(x) e^{-\epsilon x^2}$$

$$(\psi | \psi) = 0 \quad \text{EVEN FOR SOME } |\psi\rangle \neq |\phi\rangle$$

$$\epsilon > 0$$

$$\eta_{\mu\nu} x^\mu x^\nu$$

$$\eta_{\mu\nu} = \begin{pmatrix} +1 & & \\ & -1 & \\ & & -1 \end{pmatrix}$$

$$x^\mu = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

PART #6 OF OUR PROJECT

COMPLEX SCALING

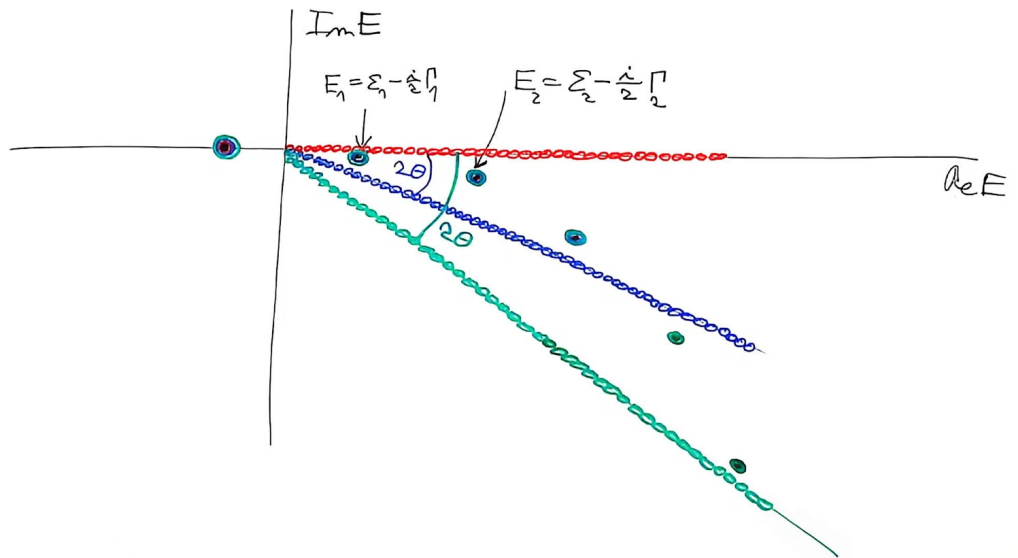
$$\hat{H}_\theta = -\frac{\hbar^2}{2m} \partial_{xx}^2 + V(x e^{i\theta})$$

$$\underline{\underline{H}} = \underline{\underline{H}}^T$$

$$\underline{\underline{H}} \neq \underline{\underline{H}}^\dagger$$

TAKE $\theta > 0$

$\theta = 0.05$
 0.10
 0.15
 0.20
 0.25



$$|T(E)|^2 = \frac{(\Gamma/2)^2}{(E - \epsilon)^2 + (\Gamma/2)^2}$$

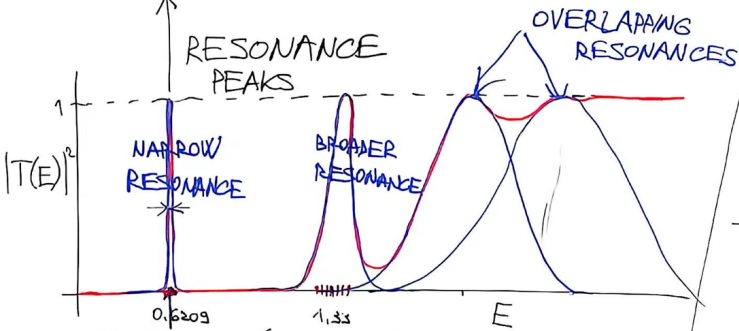
BREIT-WIGNER PROFILE

NONHERMITIAN QUANTUM MECHANICS (NHQM)

1926 TDSE

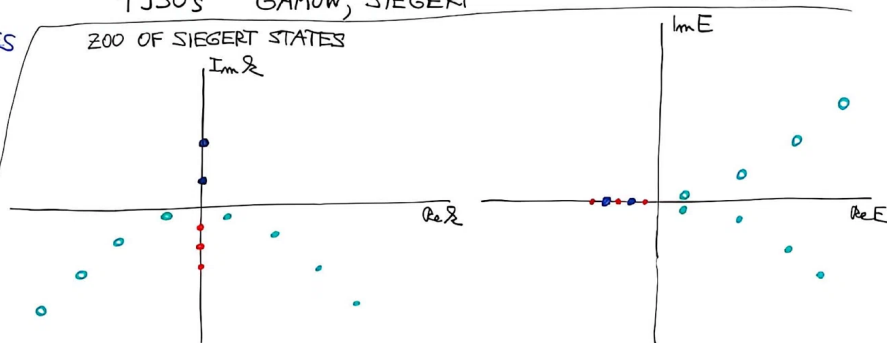
1930's GAMOW, SIEGERT

$$E = \frac{\hbar^2 k^2}{2m}$$



IN STANDARD (HERMITIAN) QM:
 RESONANCES ARE SEEN SO FAR
 AS OPAQUE FEATURES IN $|T(E)|^2$

200 OF SIEGERT STATES



● BOUND STATES $\in L^2(\mathbb{R})$

● ANTI-BOUND STATES $\notin L^2(\mathbb{R})$

● RESONANCES

● ANTI-RESONANCES

ONE CAN PROVE THAT THESE FORM
 A COMPLETE BASIS SET IN $C_0(\mathbb{R})$

ORTHONORMALITY? ONE NEEDS TO
 ESTABLISH AN
 APPROPRIATE SCALAR PRODUCT

$$E = \frac{\hbar^2}{2m} \left(k_R^2 - k_I^2 \right) + 2i k_R k_I \left(\frac{\hbar^2}{2m} \right)$$

$$k_R \neq 0$$

$$k_I < 0$$

$$-\frac{\hbar^2}{2m} \partial_{xx}^2 \psi^*(x) + V(x) \psi^*(x) = E^* \psi^*(x)$$